4 *Response generation*—Based on the combined understanding of the image and text, the model generates an appropriate response.

For example, a user could upload an image of a broken appliance with the prompt, "What is this part, and how do I fix it?" The model would identify the specific part, analyze its condition, and provide a detailed, context-aware answer.

These models can be used for various vision-related tasks, such as image captioning, visual question answering, and even generating new images based on textual prompts and visual references.

NOTE This feature is also called *multimodal AI*. It refers to AI systems that can understand and generate content using multiple modes of input, such as text, images, and sometimes audio.

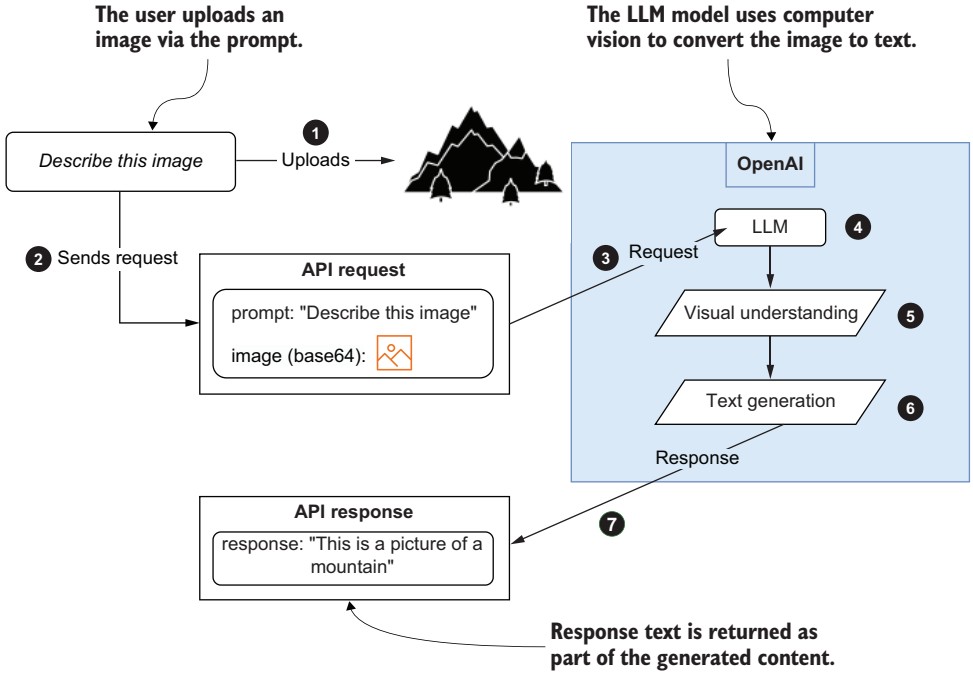


Figure 3.7 The process of sending an image alongside a text prompt to the OpenAI API and how the language model utilizes computer vision techniques to understand the image and generate a relevant text response

Figure 3.7 illustrates the process of integrating visual information into prompts. In this diagram,

1 The client application also provides an interface (file input) for the user to upload an image file.